

Motor Datasheet: WxF-50x27SL 1820 mk1

Machine Configuration

Parameter	Value	Unit
Machine Outer Diameter	48.0	mm
Machine Length / Thickness	26.3	mm
Voltage	48.0	V
Operating Temperature	-40 to 85	°C

Mass Properties

Parameter	Value	Unit
Machine Mass (active components only)	181.3	g
Rotor Mass	49.8	g
Stator Mass	131.6	g
Rotor Structure Included	No	-
Rotor Moment of Inertia	182.0	g·cm ²

Performance Metrics

Parameter	Value	Unit
Torque (continuous)	702.2	mNm
Speed	2500	rpm
Losses (continuous)	22.9	W
Supply Current (RMS)	3.96	A
Phase Resistance (per phase)	446.2	mΩ
Terminal Inductance ($L_{tt} = L_d + L_q$)	2178.5	μH
D-axis Inductance (L_d)	1144.3	μH
K_e (peak, phase)	8.51	mV/rpm
K_e (peak, line-to-line)	14.74	mV/rpm
K_t (peak)	123.2	mNm/A
Motor Constant (K_m)	153.2	mNm/√W